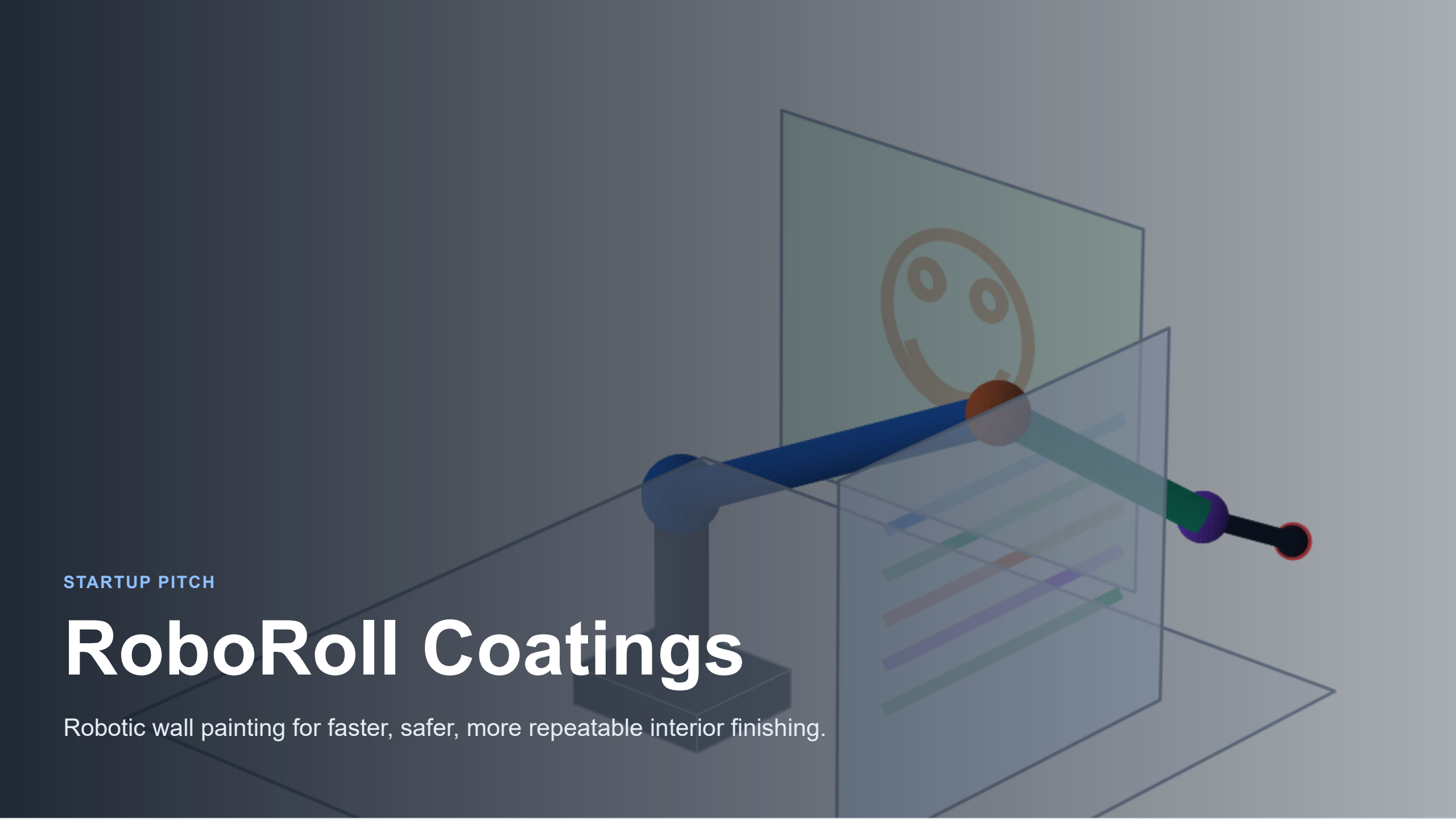


STARTUP PITCH

RoboRoll Coatings

Robotic wall painting for faster, safer, more repeatable interior finishing.

1. THE PROBLEM

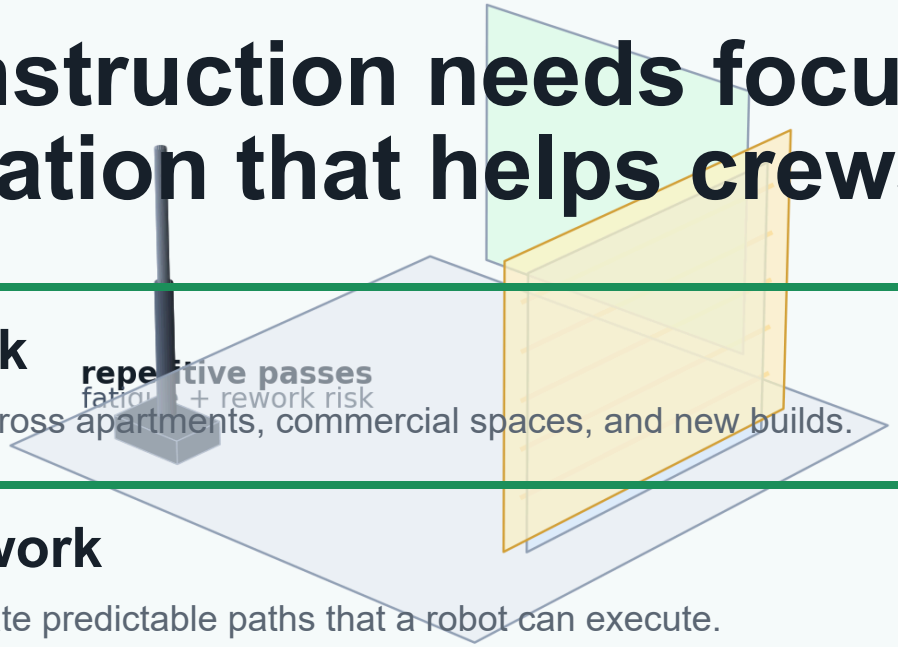
Interior painting is still a manual bottleneck.

- Repetitive wall coverage consumes crew hours.
- Quality changes with fatigue, reach, lighting, and setup consistency.
- Labor bottlenecks delay final construction turnover.

Manual wall finishing bottleneck

2. WHY IT MATTERS

Construction needs focused automation that helps crews now.



Common task

Painting shows up across apartments, commercial spaces, and new builds.

Repeatable work

Large wall areas create predictable paths that a robot can execute.

Schedule value

Fewer rework passes mean more predictable finishing timelines.

Beachhead market: multifamily apartment turns

3. THE SOLUTION

A portable 4-DOF robot that follows planned wall paths.

- Base yaw aims the arm around the room.
- Shoulder, elbow, and wrist place the nozzle on target.
- Inverse kinematics converts wall points into smooth joint motion.

Primary first customer

Multifamily property teams

Apartment turns need repeatable repainting across similar rooms, often under tight move-in schedules.

Expansion 1

Commercial tenant improvements

Large interior walls and predictable coverage paths.

Expansion 2

New-build interior finishing

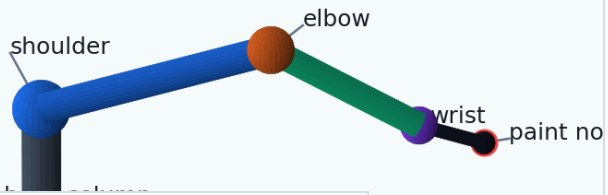
Schedule pressure near turnover after drywall completion.

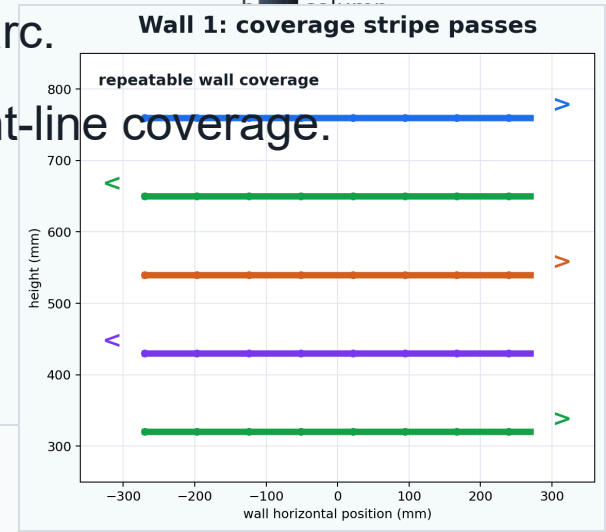
RoboRoll 4-DOF arm architecture

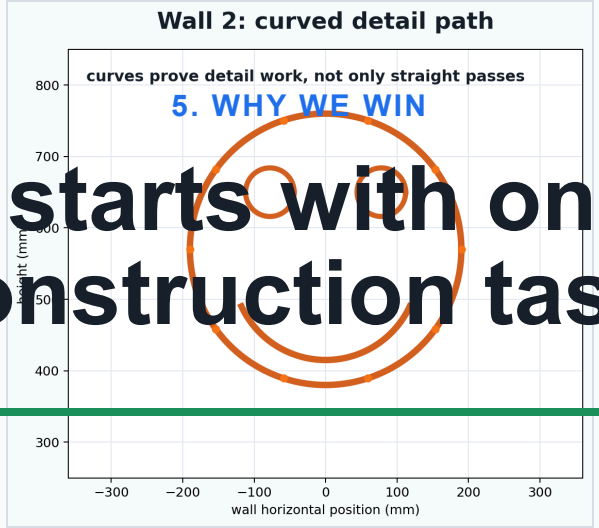
4. PRODUCT DEMO

The simulated robot already paints across two walls.

- Wall 1: five horizontal color stripes.
- Wall 2: face outline, eyes, and smile arc.
- Curved paths prove more than straight-line coverage.







RoboRoll starts with one focused construction task.

Focused wedge

Interior wall finishing is narrow enough for a first product.

Technical proof

FK, IK, trajectory generation, and dynamics are already simulated.

The ask

Seed funding turns the validated simulation into a portable prototype.

Prototype roadmap

1

Paint tool

Nozzle, flow, cleanup

2

Calibration

Wall plane sensing

3

Drywall tests

Repeatability + finish

4

Operator workflow

Setup and supervision

Thank You

ROBOROLL COATINGS

Robotic painting for the next generation of construction crews.

ADDITIONAL MATERIALS

Leave-Behind Slides

The following slides provide the fuller technical, market, and roadmap story.

A1. BEACHHEAD MARKET

Start with multifamily apartment turns.

Why first

Similar room layouts create repeatable painting paths.

Buyer pain

Turnover schedules are tight, and labor availability affects move-in dates.

Next markets

Commercial interiors and new-build finishing once setup is proven.

Beachhead market: multifamily apartment turns

A2. CUSTOMER WORKFLOW

RoboRoll helps one operator supervise repeatable wall coverage.

Primary first customer

Multifamily property teams

Apartment turns need repeatable repainting across similar rooms, often under tight move-in schedules.

Expansion

Commercial tenant improvements

Large interior walls and predictable coverage paths.

Expansion 2

New-build interior finishing

Schedule pressure near turnover after drywall completion.

1. Position

Roll the robot into the room near the target wall.

2. Calibrate

Register the wall plane and reachable painting area.

3. Paint

Execute planned coverage paths with human supervision.

4. Verify

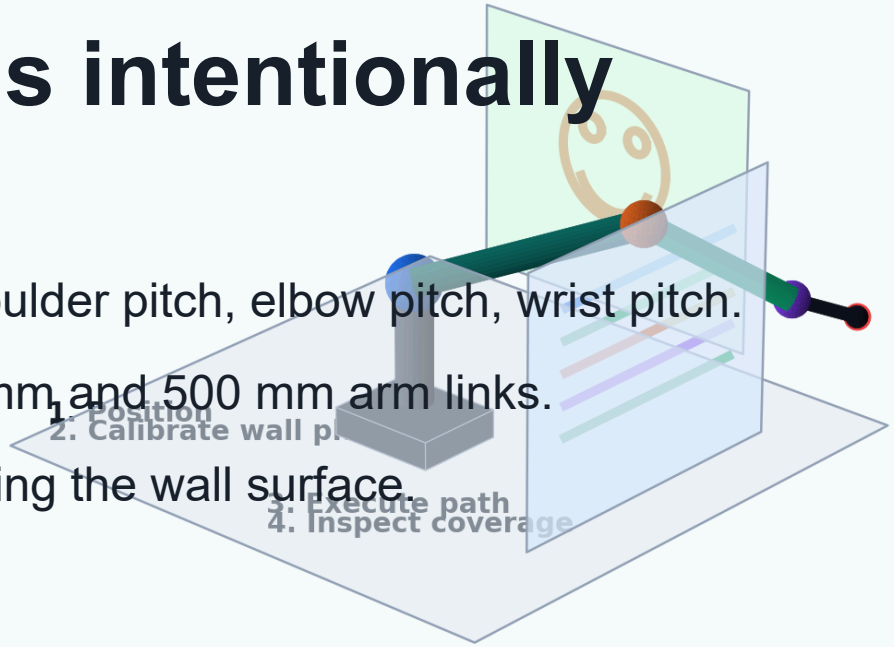
Inspect coverage, refill, and move to the next room.

Room-scale wall painting workflow

A3. ROBOT ARCHITECTURE

The first design is intentionally simple.

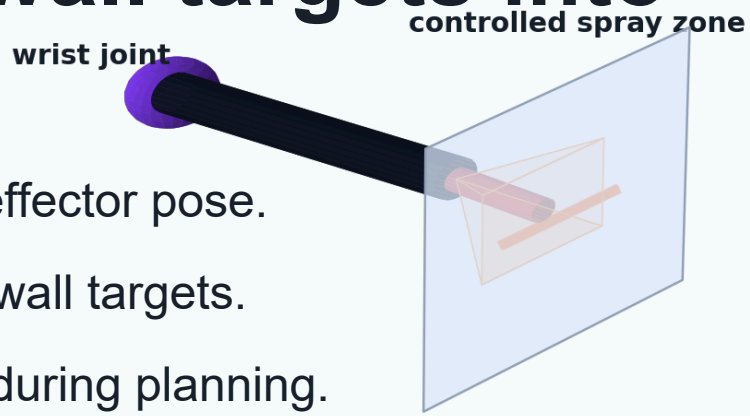
- 4 revolute joints: base yaw, shoulder pitch, elbow pitch, wrist pitch.
- 500 mm base height with 700 mm and 500 mm arm links.
- 200 mm nozzle offset for reaching the wall surface.



Paint end-effector concept

A4. KINEMATICS PROOF

The robot can turn wall targets into joint commands.



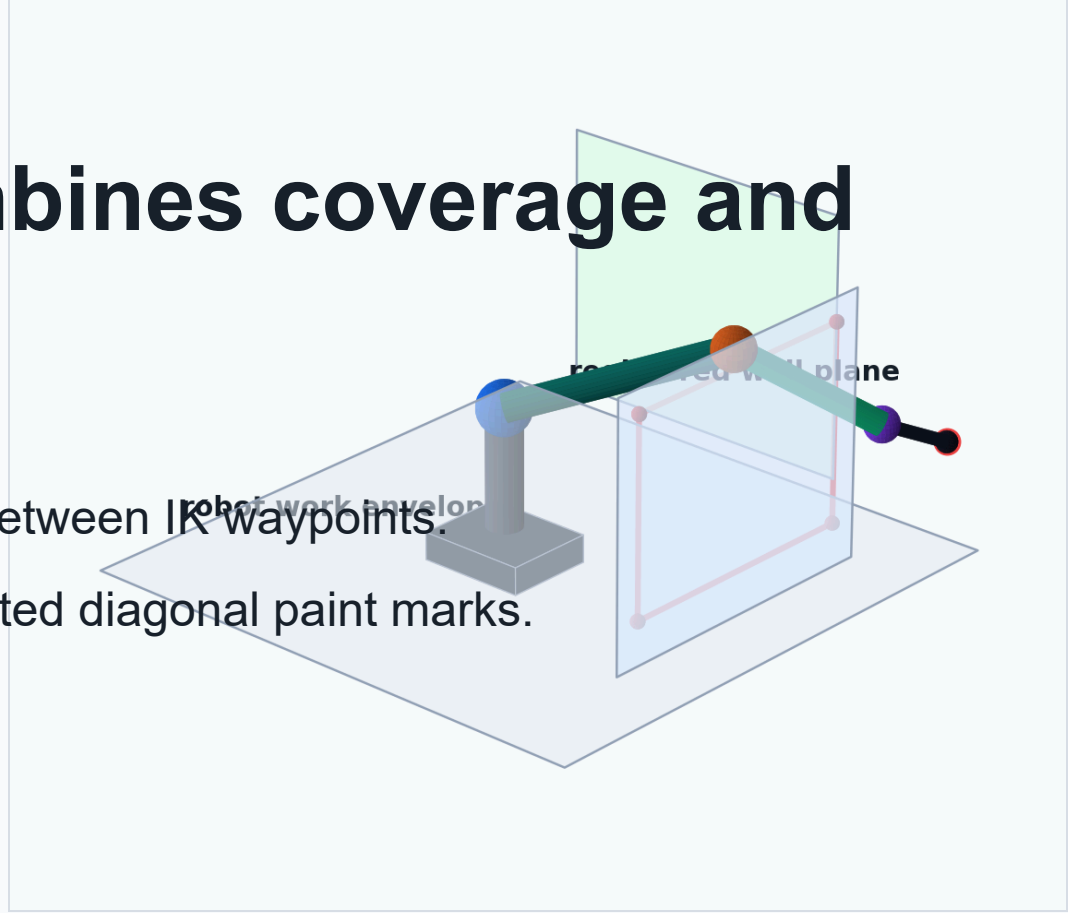
- Forward kinematics verifies the end-effector pose.
- Inverse kinematics solves reachable wall targets.
- Joint limits prevent unrealistic poses during planning.

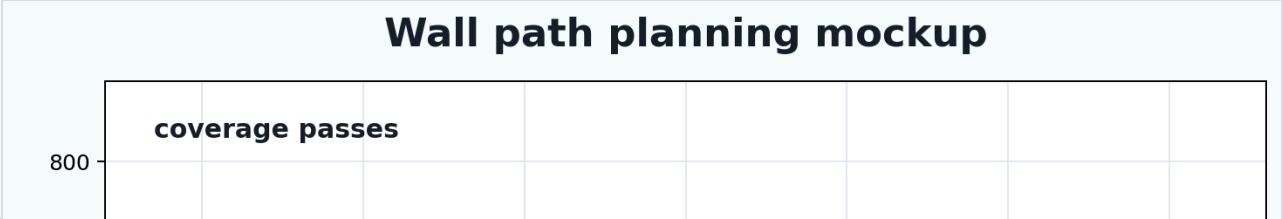
Wall calibration and work envelope

A5. DEMO PATH DETAILS

The demo combines coverage and detail work.

- 343 rendered frames.
- Smoothstep interpolation between IK waypoints.
- Lift motions prevent unwanted diagonal paint marks.

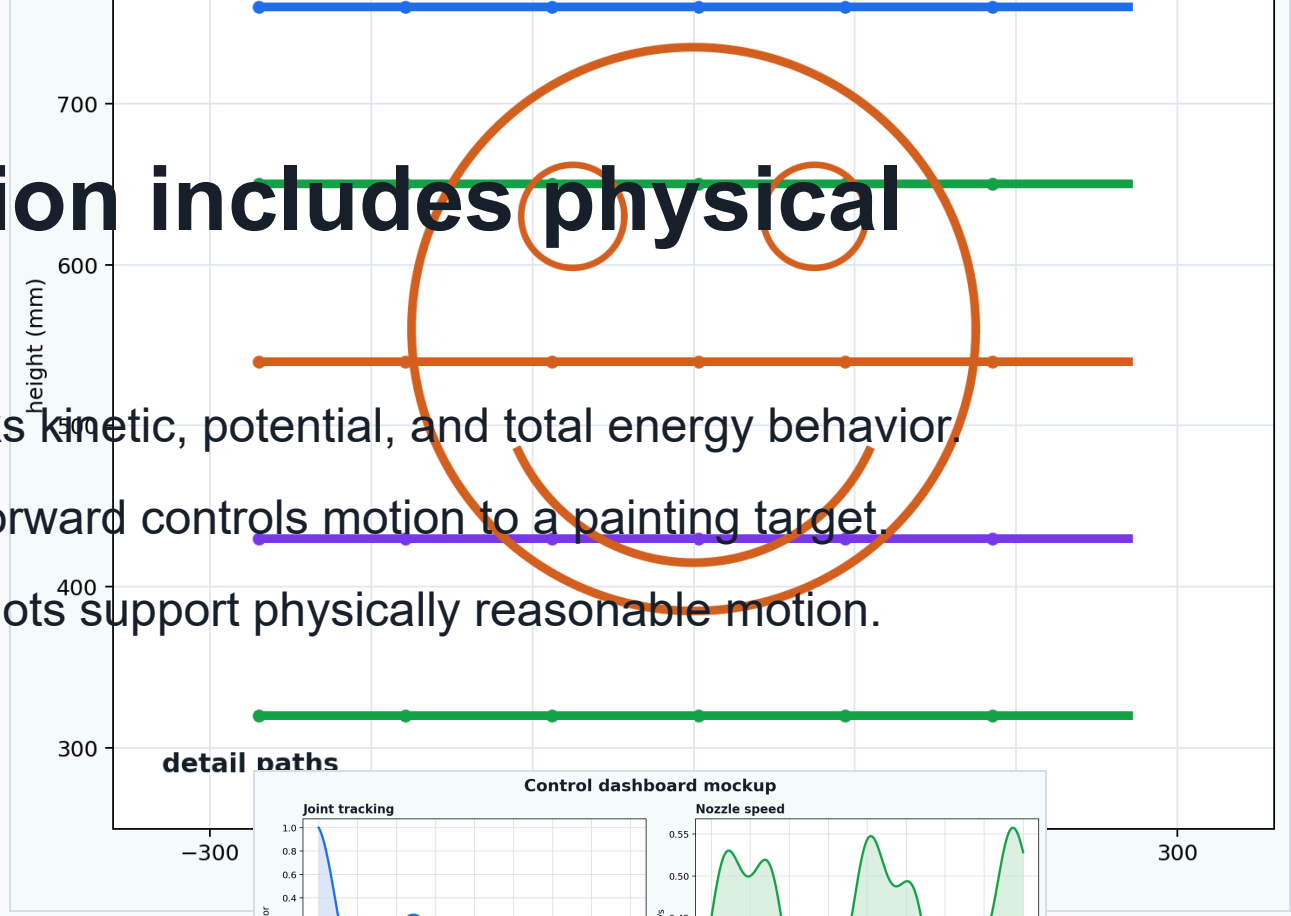


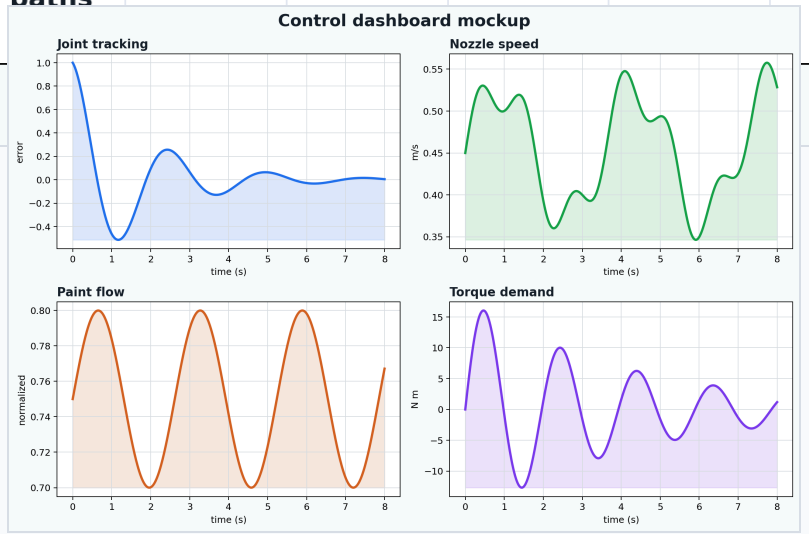


A6. DYNAMICS PROOF

The simulation includes physical reasoning.

- Passive motion checks kinetic, potential, and total energy behavior.
- PD plus gravity feedforward controls motion to a painting target.
- Torque and velocity plots support physically reasonable motion.





A7. DEVELOPMENT ROADMAP

The next milestones move from simulation to hardware.

End effector

Prototype paint delivery and nozzle control.

Calibration

Add wall sensing and room registration.

Testing

Measure repeatability on drywall panels.

Workflow

Build the operator setup process.

Prototype roadmap

A8. RISKS AND MITIGATIONS

The key risks are practical and testable.

1	2	3	4
Paint tool	**Calibration**	**Drywall tests**	**Operator workflow**
Nozzle, flow, cleanup	Wall plane sensing	Repeatability + finish	Setup and supervision

Surface quality

Test nozzle spacing, flow rate, and overlap on drywall panels.

Wall localization

Add sensing and calibration before each room run.

Setup time

Design a simple operator workflow with repeatable presets.

Safety

Limit speed, add supervised operation, and define a work envelope.

A9. FUNDING USE

Funding turns the validated simulation into a prototype.

Arm

Build a first portable mechanism.

Paint tool

Integrate nozzle, flow, and cleanup hardware.

Sensing

Add wall plane detection and calibration.

Testing

Run repeatability and finish-quality trials.

A10. CLOSING STORY

A focused wedge into construction automation.

RoboRoll begins with repeatable interior painting, proves value in controlled room-scale tasks, and expands toward broader finishing automation.